Python Class

① ~~Ask students if we can schedule a lab on 29th Friday~~ what to do?

3:45

FSM

Specification

① Propositional

$a \wedge b$

$a \rightarrow b$

② First

$\forall x$

$\exists x$

a^*

③ Predicate Logic

$\forall$ age(M, F)

q_1 q_2

④ Automata

$a \rightarrow b$

⑤ TL

CTL

$a = 1.5$

STL

MITL

FSA

M S

A A

FSM / Automata

FSA

FV

Dynamic

① Runtime Verification

② Runtime Enforcement

static

① TP

② MC

③ CT

Mech

M

Car

P_1: No coll

Syst ✗ Model ✗

execution

call()

Run a b c

event

P

Syst

abc

abc

abc

not comp

trace a b c

not

a b

Prop

a b

ab

ab

a b

Verification

Simulation

↓ Formal guarantees

Formal verification approach

↳ Theorem Proving

Static

↳ Model checking (M → FSM/Kripke, S → FSM/Logic → MC → T / f (counter example))

Mutual Exclusion

Dynamic

→ Runtime verification

↳ Runtime Enforcement

↳ block

↳ suppress

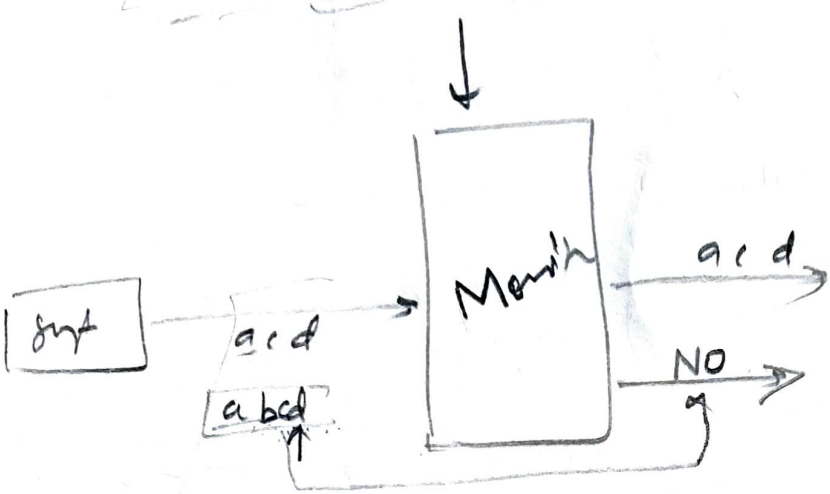
↳ insert

↳ delay/buffering.

$P_1 = \{$ a b c d ✓
a c d ✓
$\varphi_1 ; a_1$ ✓ $\}$

CTL

F.S.
S
↓
D



B. Schneider